DATABASE DESIGN & NORMALIZATION

SCENARIO 1: SMART CAMPUS HOSTEL MANAGEMENT

Entities and Attributes

Entity	Attributes
Hostel	hostel_id (PK), hostel_name
Room	room_id (PK), hostel_id (FK),capacity
Student	student_id (PK), name, email
Allocation	allocation_id (PK), student_id (FK), room_id (FK)
Complaint	complaint_id (PK), student_id (FK), description, status,created at
Maintenance_Staff	staff_id (PK), name
Complaint_Assignment	complaint_id (FK), staff_id (FK)

Relationships

- Hostel → Room = 1:N
- Student → Complaint = 1:N
- Student ↔ Room = M:N (via Allocation)
- Complaint ↔ Maintenance_Staff = M:N (via Complaint_Assignment)

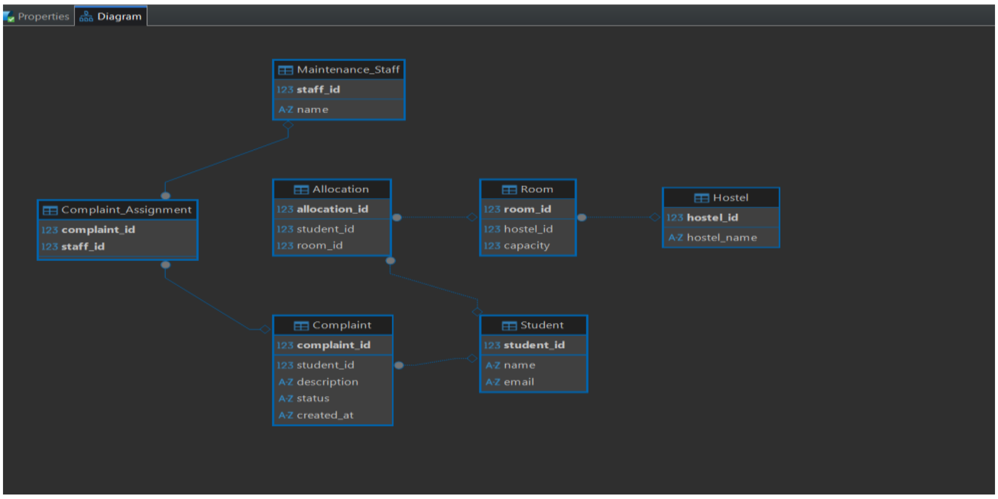
Normalization

1NF: All attributes are atomic. Repeating groups are removed.

2NF: No partial dependency; non-key attributes fully depend on the primary key.

3NF: No transitive dependency; redundancy is minimized by storing data in appropriate tables.

ER Diagram



SCENARIO 2: MINI E-COMMERCE PLATFORM

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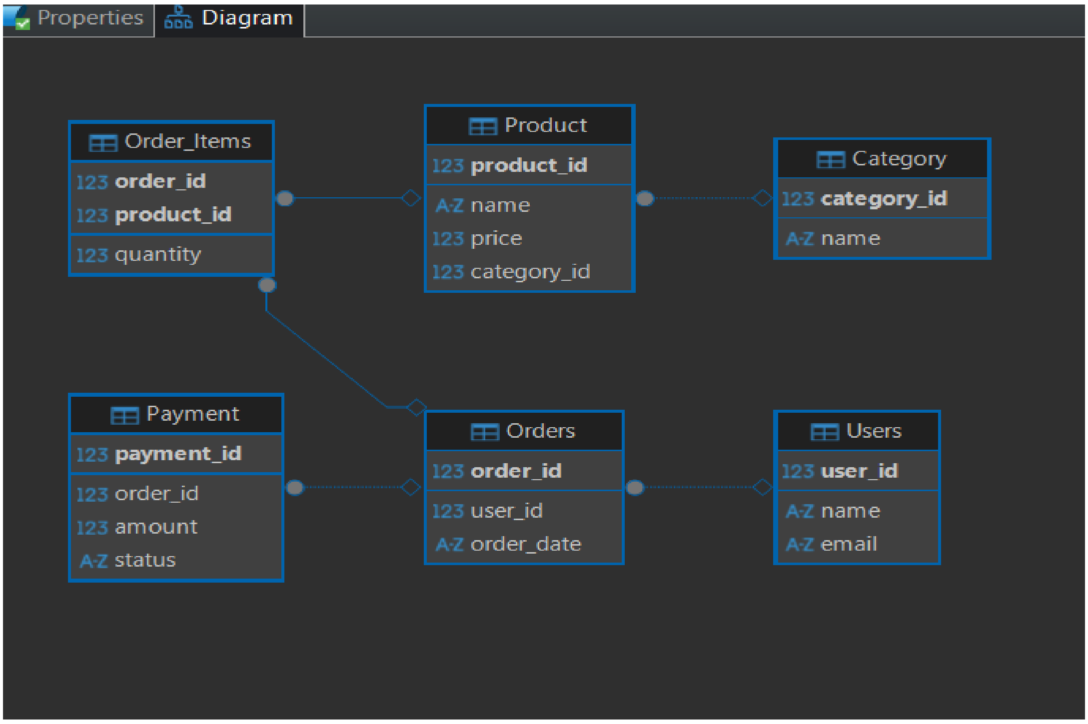
Entities and Attributes

Entity	Attributes
Users	user_id (PK), name, email
Category	category_id (PK), name
Product	product_id (PK), name, price, category_id (FK)
Orders	order_id (PK), user_id (FK), order_date
Order_Items	order_id (FK), product_id (FK), quantity
Payment	payment_id (PK), order_id (FK), amount, status

Relationships

- Category → Product = 1:N
- Users → Orders = 1:N
- Orders → Payment = 1:N
- Orders ↔ Product = M:N (via Order_Items)

ER Diagram


Properties
Diagram
Order_Items
123 order_id
123 product_id
123 quantity
Product
123 product_id
A-Z name
123 price
123 category_id
Category
123 category_id
A-Z name
Payment
123 payment_id
123 order_id
123 amount
A-Z status
Orders
123 order_id
123 user_id
A-Z order_date
Users
123 user_id
A-Z name
A-Z email